Optical CT Dosimetry

Step-by-step user guide

For medical physics students and researchers

What this guide covers

This guide walks you through every step of a complete optical CT dosimetry measurement: from initial hardware setup and calibration, through the pre-irradiation scan, irradiation of the dosimeter, post-irradiation scan, 3-D reconstruction, and final depth dose extraction. Each procedure is described in the order you will perform it in the app.

Workflow at a glance

Day 1	**Pre-irradiation scan**	Capture dark, flat, and pre-irradiation projections.
Day 1–N	**Irradiation**	Deliver the planned dose to the dosimeter (outside this app).
Day N	**Post-irradiation scan**	Capture dark, flat, and post-irradiation projections. ΔA projections are computed automatically.
After scan	**Reconstruction**	Run FBP reconstruction to obtain the 3-D attenuation volume.
After recon	**Depth dose**	Extract relative dose vs. depth and export results.

1. Background: How optical CT dosimetry works

Optical CT dosimetry uses a polymer or radiochromic gel dosimeter that darkens proportionally to the absorbed radiation dose. The app quantifies this darkening by measuring the change in optical density (ΔA) between a pre-irradiation and a post-irradiation scan.

For each projection angle, the Beer–Lambert law relates measured pixel intensity I to absorbance A:

$$A = -\log(I / I0)$$

where I0 is the flat-field (lamp-only) intensity. The dose-related signal is the difference:

$$\Delta A = A\_post - A\_pre = \log(I\_pre / I\_post)$$

Note that I0 cancels exactly, making the measurement independent of lamp-intensity drift between sessions. A set of ΔA projections at evenly-spaced angles is then reconstructed with filtered back-projection (FBP) to yield a 3-D attenuation volume, from which a depth dose profile is extracted.

Note

Dark frames (lamp OFF) are subtracted before every calculation to remove camera read-out noise and hot pixels. Always capture fresh dark and flat frames at the start of each session.

2. Hardware setup and first launch

Starting the app

1. **Launch the app.**
   Double-click the app icon or run: python3 oct_app.py
2. **The workflow dialog appears.**
   Choose the appropriate workflow for your current session (see Section 3).
3. **Check "Real camera" and "Real stepper" checkboxes in the Scan panel.**
   Leave both unchecked only for off-line testing with the camera simulator.
4. **The Lamp button in the Scan panel defaults to ON.**
   It toggles the lamp directly from the app and turns amber when the lamp is lit. During a scan the lamp is controlled automatically — turned off for dark frames, on for flat and projection captures — and the button updates to reflect the current state. The lamp is switched off automatically when the app exits.

Note

On first run the app creates a config/ folder alongside oct_app.py and stores default settings there. You can adjust defaults via File → Save current settings as defaults.

Note

Live preview: when the Pi camera is reachable, the preview panel shows a real camera image refreshed every 2 seconds. When the lamp is off the preview will be dark — this is expected. If the Pi is not reachable the panel falls back to the built-in camera simulator.

Camera settings

The camera runs with auto-exposure disabled. A fixed shutter speed is essential because Beer-Lambert law requires $A = -\log(I / I0)$ to be computed from images captured under identical conditions — any exposure change between the flat frame and a projection frame, or between the pre- and post-irradiation sessions, would introduce a spurious ΔA signal unrelated to the dosimeter.

The current shutter speed (500 ms at analogue gain 1.0) was determined empirically for correct exposure with the installed lamp and optics. If the lamp is replaced or the optical path changes, recalibrate the shutter speed by adjusting ExposureTime in camera_server.py until the flat-field image is well-exposed (pixel values in the range 180–230 on an 8-bit scale) without saturation.

3. Scan parameters — what each setting controls

These controls appear in the Scan panel (left column) and the Reconstruction panel (right column).

Parameter	Default	Description
Step size (°)	*default: 2°*	Angular increment between projections. 2° gives 180 projections over 360°, balancing scan time against reconstruction quality.
OCT stack	*default: 3*	Number of frames averaged per projection angle to reduce camera noise.
Calib stack	*default: 3*	Frames averaged for dark and flat calibration captures.
Settle (ms)	*default: 300*	Time the motor pauses after each step before the camera captures, allowing vibration to damp out.
Crop centre X (px)	*default: 995*	Horizontal pixel position of the axis of rotation. Auto-detect (button in Reconstruction panel) estimates this from the pre-scan projections.

Crop top Y (px)	*default: 270*	Top edge of the region of interest in the image. Should be just above the dosimeter top.
Crop extent (px)	*default: 700*	Side length of the square crop window. Must encompass the full dosimeter diameter.
Sample top (px)	*default: 0*	First row (within the crop) of the dosimeter sample region, used for depth dose extraction.
Sample height (px)	*default: 450*	Number of rows of the dosimeter included in the depth dose ROI.

! Warning

Crop centre X must be accurate. An off-centre axis produces blurred ring artefacts in the reconstruction. Always run Auto-detect after a pre-scan and verify the axis line overlies the rotation axis in the preview.

4. Session 1 — Pre-irradiation scan

This session is performed before the dosimeter is irradiated. It captures the baseline absorbance of the unirradiated gel. A new scan folder is created and named automatically using the current date and time.

Preparation

1. **Remove the dosimeter from any storage solution and dry the exterior.**
2. **Mount the dosimeter on the rotation stage, centred on the axis of rotation.**
3. **Ensure the lamp housing and camera are aligned and the optical path is clear.**

In the app

4. **At the workflow dialog, select "Pre-irradiation scan".**
   The app opens ready for a pre-scan session.
5. **Verify scan parameters in the Scan panel.**
   Step size, stack size, and settle time are the most commonly adjusted.
6. **Click "CAPTURE PRE SCAN".**
   The four-phase checklist activates.

Phase sequence:

(1) Dark capture	**Lamp OFF — automatic.**	Remove the dosimeter from the beam path, then click "Capture dark". The app turns the lamp off automatically before capturing dark frames to characterise read-out noise and hot pixels.
(2) Flat capture	**Lamp ON — automatic.**	Ensure no sample is in the beam, then click "Capture flat". The app turns the lamp on automatically. The camera captures the bare beam profile (I0).
(3) Pre-irradiation scan	**Lamp ON — automatic.**	Place the dosimeter on the rotation stage. Click "Begin pre-scan". The motor steps through 360° and a projection image is captured at each angle. Do not disturb the setup. The lamp remains on when the scan finishes so you can inspect the dosimeter.
(4) Post-irradiation scan	**Skipped in this session.**	This phase is automatically skipped for a pre-scan session.

7. **When the scan finishes, the axis is auto-detected from the projections.**
   The crop centre X spinbox updates automatically. Check that the vertical axis line in the preview overlies the physical rotation axis.
8. **Irradiate the dosimeter.**
   Remove the dosimeter, deliver the planned dose, then store it undisturbed until the post-irradiation scan. Note the scan folder name — you will select it in Session 2.

Note

The app saves projections to scans/<name>/pre/ and calibration frames to scans/<name>/calibration/. Do not rename or move this folder.

5. Session 2 — Post-irradiation scan

This session is performed after irradiation, ideally within the dosimeter's stable post-irradiation window (consult the gel protocol for timing requirements). The post-scan is stored alongside the pre-scan data in the same folder. ΔA projections are computed automatically on completion.

In the app

1. **At the workflow dialog, select "Post-irradiation scan".**
   A scan selector appears.
2. **Select the pre-scan folder for this dosimeter from the list.**
   Only scans with a completed pre/ folder are shown.
3. **Click "Continue".**
   The app opens with scan mode set to Post-irradiation.
4. **Click "CAPTURE POST SCAN" and follow the same four-phase sequence.**
   Phases (1) and (2) (dark and flat) are re-captured — fresh calibration frames correct for any lamp drift since the pre-scan. Phase (3) is skipped. Phase (4) captures the post-irradiation projections. The lamp remains on after the scan completes.
5. **On completion, ΔA = A_post − A_pre is computed for every projection.**
   Frames are paired by the rotation angle in the filename, so a pre and post scan that used different angle settings cannot be silently mismatched — the log reports any unpaired angles. Pixels too dark to measure reliably (below 10 counts in either frame) are masked and reported as a percentage. Results are saved to scans/<name>/subtracted/ as 16-bit PNG files.

! Warning

Mount the dosimeter in the same orientation as the pre-scan. Any rotation of the dosimeter about the vertical axis between sessions will degrade the reconstruction quality.

Note

Tip: mark the top of the dosimeter with a small permanent marker dot before the pre-scan. Use it to align the securing screws and the "front" of the mount consistently each time the dosimeter is inserted.

Note

The flat-field term I0 cancels mathematically in ΔA = log(I_pre/I_post), so small lamp-intensity changes between sessions do not bias the result. However, large changes (lamp replacement, optical realignment) should be avoided.

6. Reconstruction

Reconstruction converts the ΔA projection set into a 3-D attenuation volume using filtered back-projection (FBP) with a Hann filter. The volume is stored as a NumPy array and cached; re-running reconstruction with the same crop parameters reloads the cache unless "Force new volume" is checked.

Steps

1. **In the Reconstruction panel (right column), select the scan.**
   Only scans with a subtracted/ folder are listed.
2. **Verify the crop and sample parameters.**
   If a previous reconstruction exists, its parameters are loaded automatically. Adjust if needed — changing any crop spinbox will automatically tick "Force new volume".
3. **Click "RECONSTRUCT".**
   Progress is shown in the progress bar. FBP runs in parallel across CPU cores; a typical 700×700 crop takes 1–3 minutes depending on hardware.
4. **Review the depth dose plot.**
   The relative dose vs. depth curve appears in the plot panel. Use the Relative / Absolute (OD) toggle to switch the Y axis. Hover the cursor over the plot to read off interpolated values.
5. **The dose centroid is detected automatically.**
   The app finds the brightest 20 % of slices in the sample ROI and computes the weighted centroid of the dose distribution. This handles beams that are off-axis by up to ~6 mm. The centroid offset is logged to the status panel and recorded in recon_config.json as dose_centroid_x_mm and dose_centroid_z_mm. X is the left/right displacement as seen in the camera image (negative = left of axis). Z is the front/back displacement along the camera line of sight (negative = closer to the camera than the rotation axis).
6. **Review the sanity-check visualisation.**
   Saved to scans/<name>/reconstruct/sanity_check.png. See Section 7 for how to interpret it.

Output files

reconstruct/attenuation_volume.npy	Cached 3-D FBP volume (float32, Y×Z×X).
reconstruct/example_cropped.png	First projection after cropping — quick crop check.
reconstruct/sanity_check.png	4-panel diagnostic figure (see Section 7).
depth-dose/depth_dose.png	Depth dose plot (relative dose vs. mm).
depth-dose/depth_dose.xlsx	Tabulated depth_mm, rel_dose, dose_signal, OD.
depth-dose/recon_config.json	Exact reconstruction parameters and the auto-detected dose centroid offset (mm) from the axis of rotation.
dose-profiles/	Per-slice radial dose profiles (every ~5% of depth).

Spreadsheet columns — depth_dose.xlsx

depth_mm	Depth position within the sample region in millimetres, starting at 0 at the top of the sample ROI and increasing at 0.1 mm per slice. This is the x-axis of the depth dose plot.
optical_density_depth	Raw mean ΔA (change in optical density) at each depth slice, averaged over the central dose ROI cylinder. Direct output of the FBP reconstruction with no further processing; values are in OD units.
dose_signal	Baseline-corrected ΔA: the mean of the top and bottom ~25 % of the sample

	column is used as a background baseline and subtracted, then negative values are clipped to zero. Removes any DC offset from residual non-dose absorption. Shown when the "Absolute (OD)" toggle is selected in the plot.
rel_dose	dose_signal normalised to its own maximum, giving a 0–1 relative dose curve. This is the default plot view and the most useful column for comparing profiles across different measurements or irradiation conditions.

Note

The Excel file contains raw (unsmoothed) values. The EMA smoothing shown in the depth dose plot is for display only and is not written to the spreadsheet.

Spreadsheet columns — dose_profiles.xlsx

Two sheets: relative_dose and optical_density. Both have the same layout: rows are lateral position in mm (pos_mm), columns are sampled depth slices labelled by their depth (e.g. 3.50mm). Approximately 20 evenly-spaced depths are sampled, covering the full reconstructed volume.

Scan folder directory structure

pre/	Raw intensity projection PNGs captured before irradiation. One file per angle, named img_NNNN_DDD_deg.png.
post/	Raw intensity projection PNGs captured after irradiation. Same naming convention as pre/.
subtracted/	ΔA = A_post − A_pre projections encoded as 16-bit PNG in offset binary (value = (ΔA + 1) × 65535/5, covering −1 to +4 OD, so negative ΔA is kept). encoding.json records the format. Input to the FBP reconstruction.
calibration/	Dark and flat calibration frames (dark.npy, flat.npy) captured at scan time. Bundled here so the correct calibration is always available alongside the projection data, regardless of when the scan is exported.
reconstruct/	attenuation_volume.npy — cached FBP volume (float32, depth × Z × X). example_cropped.png — first projection after cropping, for a quick crop check. sanity_check.png — 4-panel diagnostic figure.
depth-dose/	depth_dose.png — relative dose vs. depth plot. depth_dose.xlsx — tabulated depth dose data (see columns above). recon_config.json — all reconstruction parameters and the detected dose centroid offset from the axis of rotation.
dose-profiles/	Per-slice radial dose profile PNGs (~20 evenly-spaced depths). dose_profiles.xlsx — raw profile data: two sheets (relative_dose, optical_density), rows = lateral position (mm), columns = depth slice.

7. Interpreting the sanity-check figure

Sinogram (top-left)	One row of the projection data at the mid-sample depth, plotted as angle (°) vs. column (px). In a good scan with real dose signal you will see sinusoidal arcs tracing how each absorbing feature sweeps across the detector. A flat, featureless sinogram means the signal is below noise. Fixed vertical bands indicate a stationary feature (e.g. container wall) that did not cancel in the ΔA subtraction.
Axial slice (top-right)	Horizontal cross-section (XZ plane) through the reconstructed volume at the mid-sample depth. A well-aligned scan shows a roughly circular dose distribution centred on the axis. A bright ring at the boundary is a container-wall artefact. The small teal circle marks the 10 px ROI used for depth dose extraction.
Sagittal slice (bottom-left)	Vertical slice (YZ plane) through the centre of the volume. Depth (mm) is on the vertical axis; lateral position (mm) is on the horizontal axis. Teal dashed lines bound the sample ROI; the white dotted line marks the exact depth row shown in the sinogram panel; the vertical teal strip marks the lateral dose ROI. A horizontal brightness gradient within the ROI is the depth dose you are trying to measure.
Depth dose (bottom-right)	Smoothed relative dose vs. depth curve for this reconstruction, included to allow direct comparison with the sagittal slice and confirm that the ROI boundaries capture the correct region.

8. Exporting a scan

Use File → Export scan (or the "Export" workflow) to copy a completed scan folder to a chosen destination (e.g. a USB drive). The exported folder contains the full scan directory tree, including:

- Raw pre and post projection PNGs
- ΔA (subtracted) projection PNGs
- Calibration frames (dark.npy and flat.npy captured at scan time)
- scan_meta.json — acquisition parameters (step size, stack depth, date/time)
- recon_config.json — reconstruction parameters
- Depth dose plot and Excel table
- Sanity-check visualisation

9. Troubleshooting

Blurred or ring artefacts in the axial slice

Crop centre X is incorrect. Click Auto-detect in the Reconstruction panel after selecting the scan, or adjust manually until the axis line in the preview overlies the physical rotation axis. Tick "Force new volume" before reconstructing.

Depth dose is flat or all noise

The ΔA signal is below the noise floor. Check that the correct pre-scan was selected during the post-scan session. Inspect example_cropped.png in the reconstruct/ folder to confirm projections contain visible contrast.

"Force new volume" is ticked unexpectedly

Any change to the crop spinboxes automatically ticks this box to prevent stale cached volumes. This is intentional.

Phase button does not become active

The previous phase is still running or the worker is waiting. Check the log panel (bottom right) for error messages.

Axis auto-detect gives a wrong value

Auto-detect looks for the nozzle holder — a dark vertical band in the averaged projections. If the band contrast is below 5 % it returns the previous value unchanged. Verify lamp alignment and ensure the nozzle is within the crop window.

Excel export fails

pandas is not installed in the active environment. The depth dose data is still available in the depth_dose.xlsx file once pandas is available, or use the depth dose plot PNG as an alternative.

Live preview is dark or shows the simulator animation

If the lamp is off, the preview will be black — click the Lamp button to turn it on. If the preview shows the animated simulator pattern rather than a real camera image, the app cannot reach the Pi camera server: check that the Pi is powered on, that "Real camera" is ticked, and that the Pi IP address was entered correctly at startup.